

ENEL700 Tutorial 2 - Solutions

1. What is the voltage gain of an amplifier with an output of 1.5V and an input of 30μV?

$$G_v = \frac{1.5V}{30\mu V} = 5 \times 10^4 = 20 \log_{10}(5 \times 10^4) \text{ dB} = 93.98 \text{ dB}$$

2. A piece of communication equipment has two stages of amplification with voltage gains of 40 and 60 and two loss stages with attenuation factors of 0.03 and 0.075. The output voltage is 2.2V. What are the overall voltage gain (or attenuation) and the input voltage?

$$G = G_1 \times G_2 \times A_1 \times A_2 = 40 \times 60 \times 0.03 \times 0.075 = 5.4$$

$$V_{in} = V_{out} / G = 2.2V / 5.4 = 0.407V$$

3. A power amplifier has a power gain of 55dB. The input power is 600mW. What is the output power?

$$P_{out} = P_{in} G_p = 600mW \times 10^{\frac{55}{10}} = 189736.6W$$

4. What is the value of inductance that will resonate with an 80pF capacitor at 18MHz?

$$f_r = \frac{1}{2\pi\sqrt{LC}} \Rightarrow L = \frac{1}{(2\pi f_r)^2 C} = \frac{1}{(2 \times \pi \times 18 \times 10^6)^2 \times 80 \times 10^{-12}} H = 0.978 \mu H$$

5. Sketch the amplitude spectra of the following signal –

$$v(t) = 5 + 50\sin(4 \times 10^6 t + 45^\circ) - 45\sin(8 \times 10^6 t) + 30\sin(1.2 \times 10^5 t - 60^\circ)$$

and determine the normalised average power contained in the waveform.

This signal contains the following frequency components:

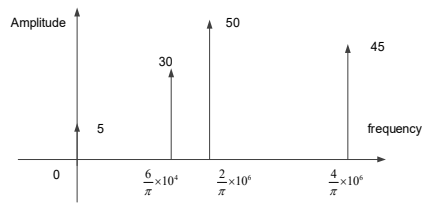
(1) DC signal, Amplitude $A_0=5$

(2) $f_1 = \frac{2}{\pi} \times 10^6 \text{ Hz}$, Amplitude $A_1=50$

(3) $f_2 = \frac{4}{\pi} \times 10^6 \text{ Hz}$, Amplitude $A_2=45$

(4) $f_3 = \frac{6}{\pi} \times 10^4 \text{ Hz}$, Amplitude $A_3=30$

The frequency spectrum is shown below:



Frequency Spectrum of the given signal

Then, the average power (normalised to 1- Ω load) can be calculated as:

$$P_{average} = (5)^2 + \left(\frac{30}{\sqrt{2}}\right)^2 + \left(\frac{50}{\sqrt{2}}\right)^2 + \left(\frac{45}{\sqrt{2}}\right)^2 = 2737.5$$